Stanley Nelavala

Full Stack Software Engineer
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Huntington, IN | Open to relocate anywhere

Summary

Full stack engineer at a 10-person hospital operations startup: FastAPI and PostgreSQL backends, React and TypeScript web apps, and a React Native app released to iOS and Android through separate pipelines. Cut a deep catalog query from 285 ms to 0.20 ms and raised mobile test coverage from 14% to 93%. M.S. in Computer Science (4.0 GPA). AWS Certified Solutions Architect, Associate.

Experience

Software Development Engineer Nov 2025 – Present
Mass Technologies, team of 10 (RxApps360 hospital platform, pre-launch; numbers from dev and test) *Huntington, IN*

- Built RxDesign, the hospital design service, backend and web (FastAPI, React, TypeScript), with a deterministic, rule-based layout engine that places rooms on a 0.5 ft grid with doors, exits and flow lines; 220+ offline tests on the API.
- Built the RxAsset iOS and Android app (React Native, Expo) with scan-to-search and an on-device voice assistant (no cloud call).
- Set up separate iOS and Android release pipelines in GitHub Actions with automatic TestFlight submission, and raised mobile statement coverage from 14% to 93% (582 tests).
- Replaced offset paging with keyset paging on the catalog device list: on a 328,000-device dev database, the query for row 150,000 fell from 285 ms to 0.20 ms. Search pages never skip or repeat rows, tested with rows inserted, deleted and tied between pages.
- Cut the planning Items page from 3 API calls and 162 KB to 2 calls and 37 KB with typeahead search and cursor paging.
- Designed, built and integrated into every platform module the signed service-to-service API (M2M): each call proves which service and tenant sent it. After a security audit, hardened it with a Redis rate limiter and replay protection.

Adjunct Faculty (part-time) Jan 2026 – Aug 2026
The College of St. Scholastica *Duluth, MN*

- Taught Programming II (Java) and Machine Learning for Data Science.

Graduate Research Assistant and Teaching Assistant Aug 2023 – Aug 2025
University of Minnesota Duluth *Duluth, MN*

- Built a pipeline (Python, Node.js, MySQL, FFmpeg, PaddleOCR) that collected and analyzed 1M+ TikTok videos from 300k+ authors for thesis research on misinformation, then grouped themes with BERTopic; presented at the UMD CS graduate poster session. Led labs and office hours for Python, NLP and Parallel Programming.

Projects (public code, tests)

Customer Integration Console | *TypeScript, Next.js, Fastify, PostgreSQL, OIDC, Trivy* GitHub

- OIDC discovery with PKCE against a mock IdP and client-credentials scopes for machine clients; an authorization matrix test over every route and role and 10 negative token tests.
- Envelope encryption (AES-256-GCM) of source credentials with key rotation, a hash-chained audit log with tamper detection, CSP and security headers; 256 tests plus 7 browser tests.

Ledger Ops Console and QA Suite | *React, TypeScript, Playwright, axe, Cucumber, Selenium, JMeter* GitHub

- React 19 and TypeScript ops console over read-only endpoints that share one keyset paging contract: 32 Vitest and 13 Playwright tests, 6 axe tests with zero WCAG 2.1 A and AA violations.

Agent Integration Platform | *Python, FastAPI, LangGraph, MCP, PostgreSQL (pgvector), Redis, Terraform* GitHub

- LangGraph case workflow with typed state, a Postgres checkpointer, human approval that resumes after restart, and step and token budgets; 62 deterministic replay evals, prompt-injection tests, PII redaction and an append-only audit log.

Skills

Languages: Python, TypeScript, JavaScript, SQL, Java, C#, Go, Rust
Frontend and mobile: React, Next.js, React Native, Expo, Sass, Vite, accessibility (axe, WCAG 2.1 A and AA)
Backend and data: FastAPI, Fastify, Spring Boot, ASP.NET Core, REST, PostgreSQL, Redis, migrations (Alembic, Flyway)
Cloud and delivery: AWS (ECS, RDS, S3, SQS), Docker, GitHub Actions, Terraform, OpenTelemetry
Testing: pytest, Jest, Vitest, Playwright, k6, Cucumber, Selenium

Education and Certification

M.S. Computer Science, University of Minnesota Duluth, 2023 – 2025, GPA 4.0, Graduate Summer Research Fellowship 2024 and 2025
B.Tech Computer Science and Engineering, IIIT D&M Kancheepuram, 2019 – 2023
AWS Certified Solutions Architect, Associate, Feb 2025 – Feb 2028 (verifiable on Credly)